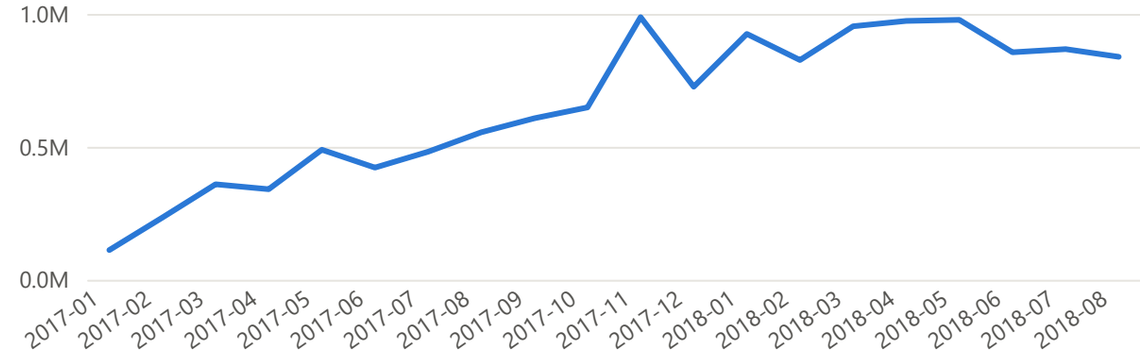
Why late deliveries cost Olist its reviews

96,203 delivered orders · Jan 2017 – Aug 2018 · Late = delivered after the promised date · Source: Olist (Kaggle)

Month: All | Region: All | State: All

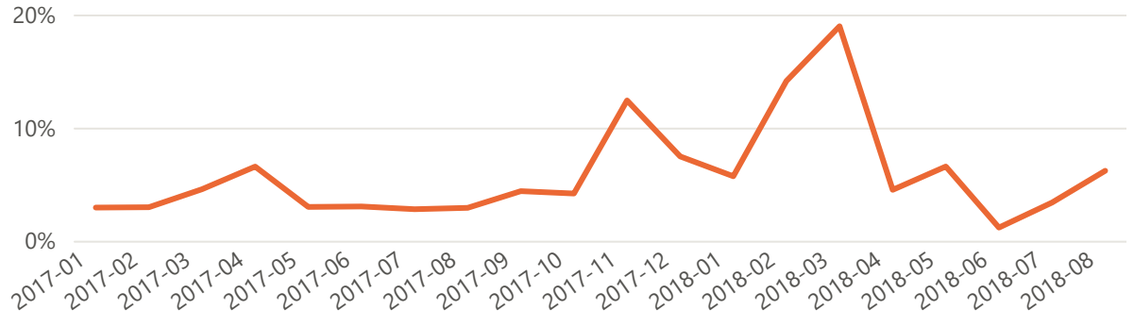
Orders	Revenue (R$)	Avg order value (R$)	Orders delivered late	Avg review score	Customers who reorder
96,203	13M	137.00	6.8%	4.14	3.00%

Revenue by month (R$)



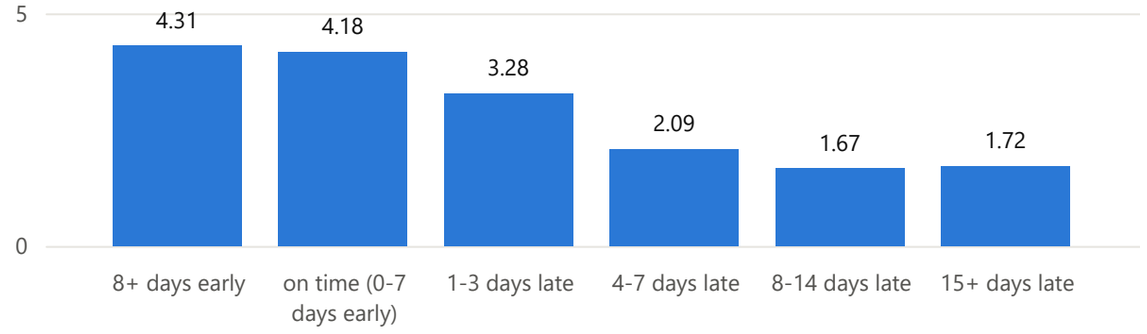
Share of orders delivered late, by month

Spikes: Nov 2017 (Black Friday month, 12%) and Feb–Mar 2018 (14–19%)



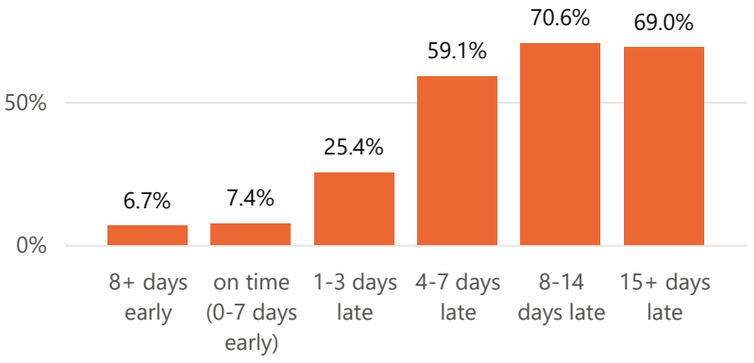
Customers forgive slow. They don't forgive late.

Average review score by delivery vs the promised date



Share of orders rated 1 star

by delivery vs the promised date



of all 1-star reviews come from the 6.8% of orders that arrive late

36.6%

Customers grade the promise, not the speed

Same delivery time, different promise: orders that met the estimated date score 0.5–1.8 stars higher

Month: All | Region: All | State: All

Avg score: met the promise

4.28

Avg score: missed the promise

2.26

1-star rate: met the promise

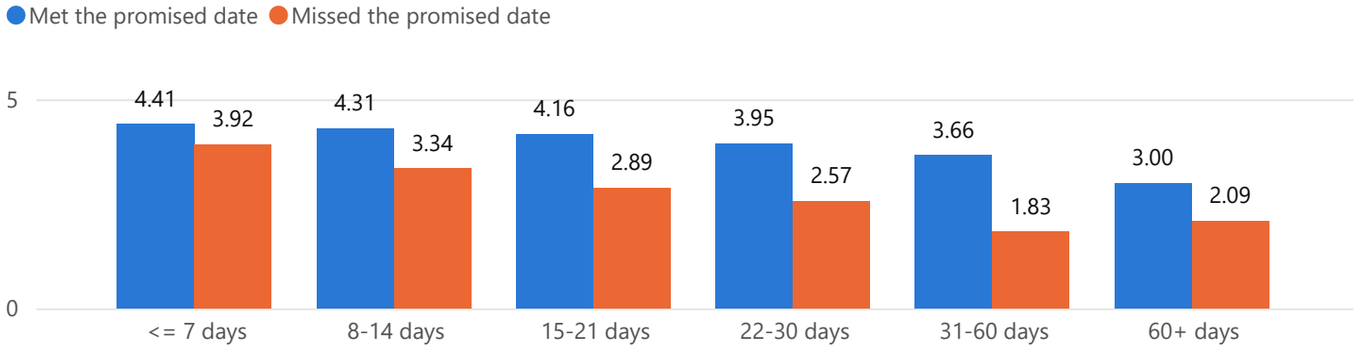
6.8%

1-star rate: missed the promise

54.2%

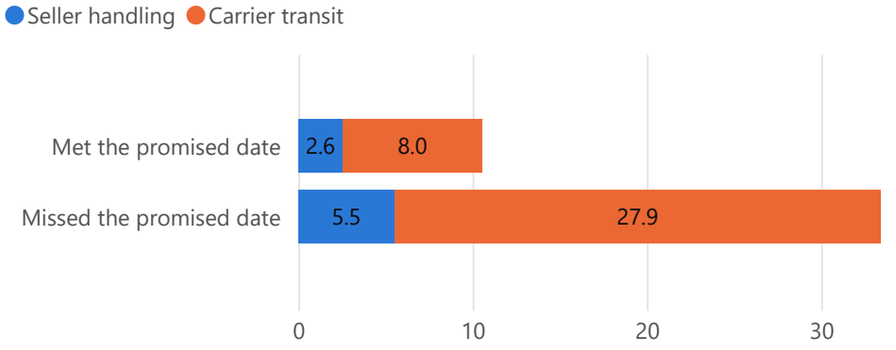
Average review score by actual delivery time, split by whether the promise was met

purchase → delivered, in days



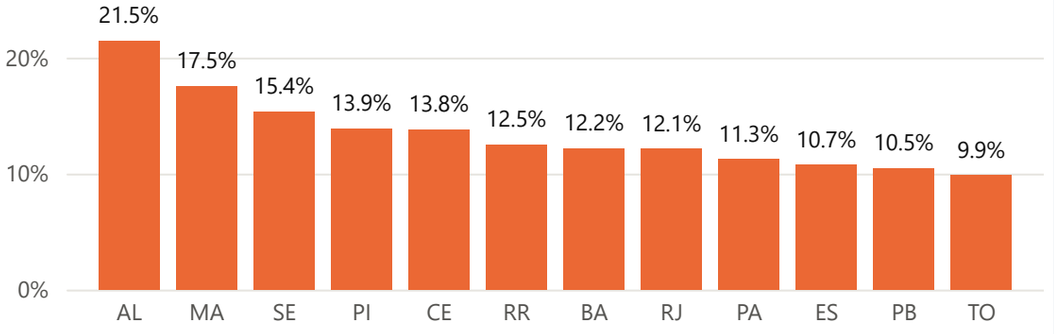
Where the time goes: seller handling vs carrier transit (days)

late orders lose ~20 extra days with the carrier, not the seller



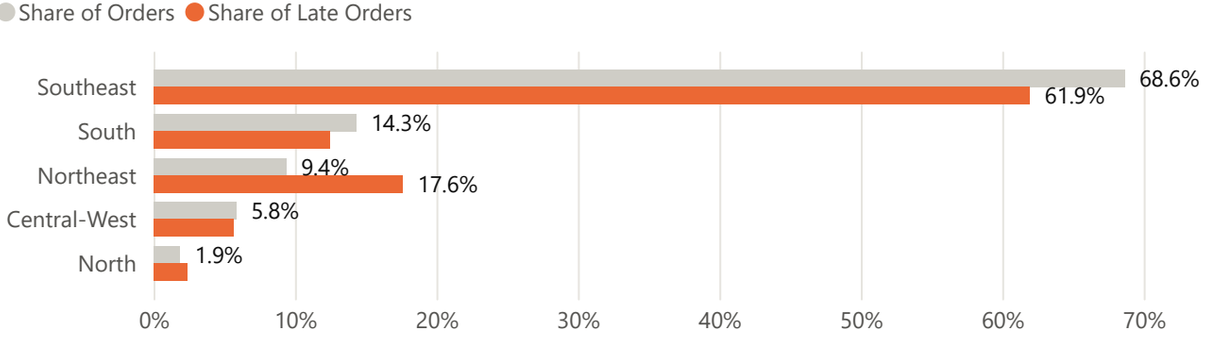
The 12 states with the highest late-delivery rate

national average 6.8%; São Paulo 4.5%



Share of all orders vs share of all late orders, by region

the Northeast and Rio de Janeiro (Southeast) are over-represented among late orders



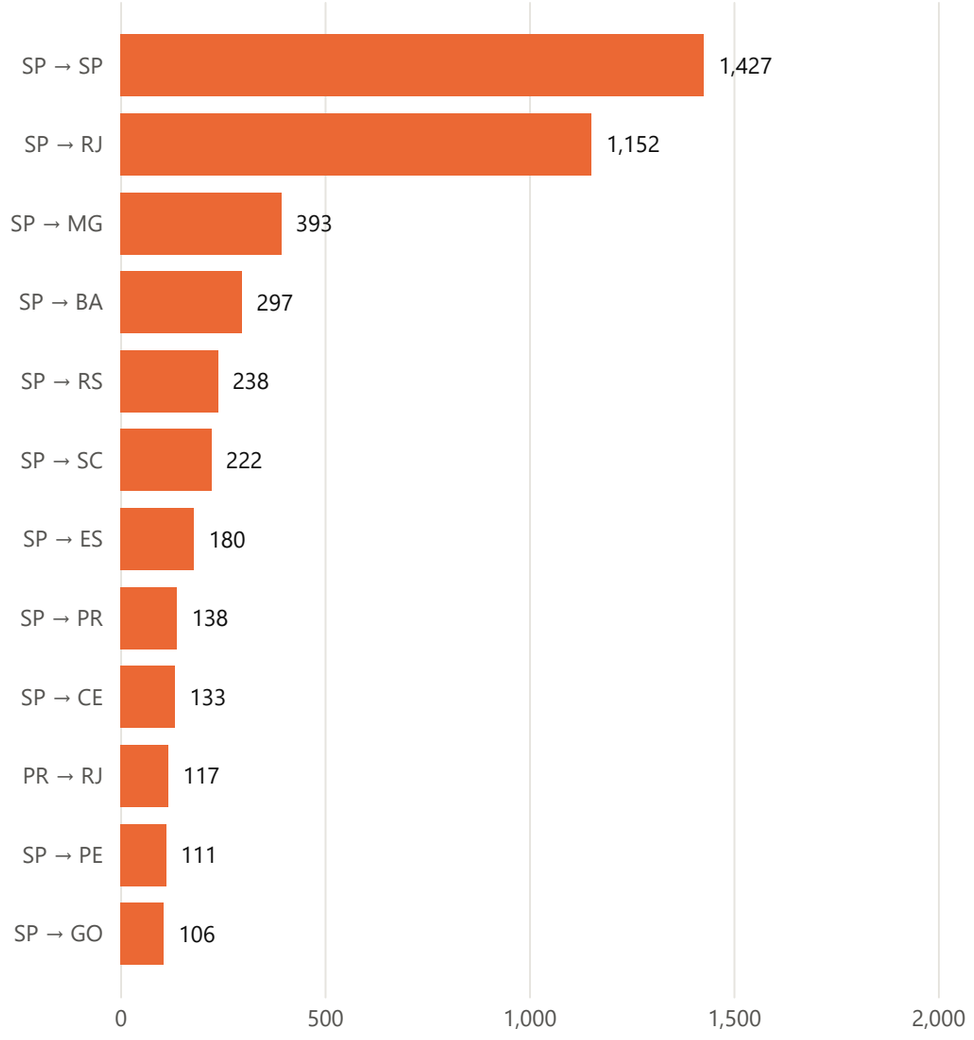
Twelve routes produce 69% of late orders — and it's the carrier leg

In 72% of late orders the seller shipped on time. SP → RJ alone is 18% of all late orders.

Month: All | Region: All | State: All

Late orders by route: seller state → customer state

the 12 routes with the most late orders = 69% of all late orders



Late rate by route: seller state (rows) × customer state (columns)

5 biggest seller states × 12 biggest customer states; darker = later

Seller state	BA	CE	DF	ES	GO	MG	PE	PR	RJ	RS	SC	SP
MG	9.8%	12.3%	3.2%	4.7%	2.7%	2.5%	7.3%	1.1%	8.3%	4.4%	6.3%	3.6%
PR	14.7%	14.3%	2.7%	8.4%	4.7%	3.9%	10.5%	2.7%	12.1%	4.1%	5.4%	3.0%
RJ	11.1%	21.2%	8.2%	2.0%	5.7%	6.1%	10.0%	8.5%	4.8%	8.6%	9.9%	7.9%
SC	11.5%	18.2%		3.8%	5.4%	3.5%	5.3%	2.5%	9.2%	4.8%	4.3%	3.7%
SP	12.9%	13.8%	6.5%	12.3%	7.5%	5.3%	9.8%	4.4%	14.2%	6.6%	9.6%	4.7%

Late orders

6,531

Avg days late, when late

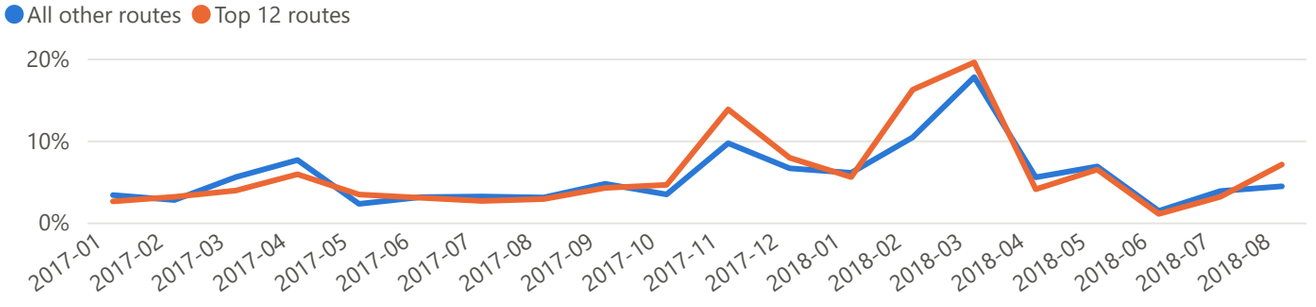
10.6

of late orders: the seller shipped on time

72.3%

Share of orders delivered late, by month: the 12 routes vs everywhere else

the Black Friday and Feb–Mar 2018 spikes hit the 12 routes hardest



Who to call on Monday: sellers to fix, categories to protect

Sellers with 100+ orders ranked by late rate · top categories by revenue (hover a bar for its late rate and review score)

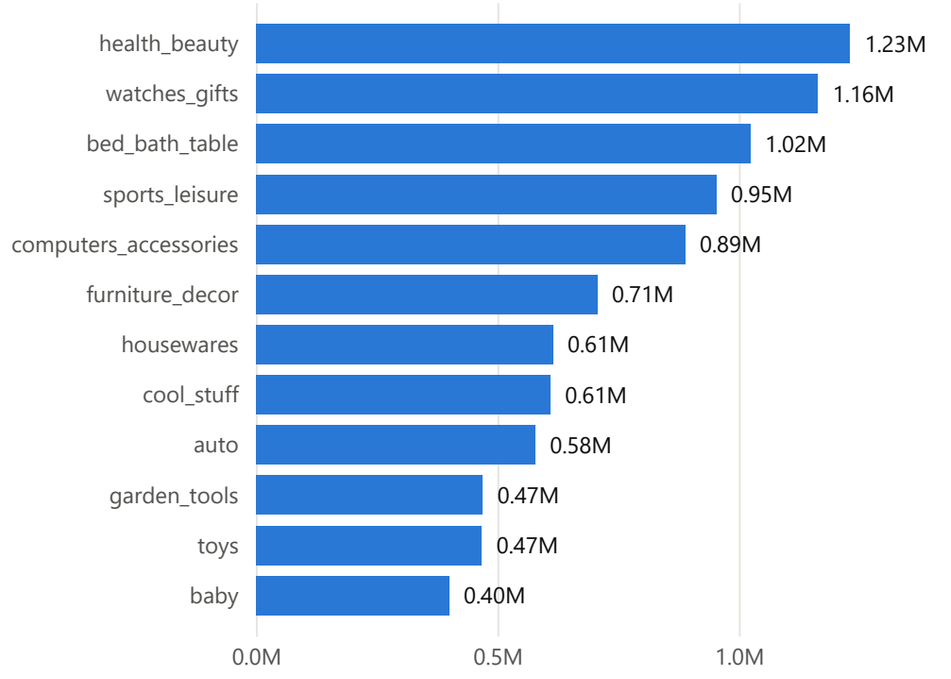
Month: All | Region: All | State: All

Seller watchlist: sellers with 100+ orders, worst late rate first

Seller ID	State	City	Orders	Late rate	Shipped late	Avg score
06a2c3af7b3aee5d69171b0e14f0ee...	MA	sao luis	389	19.0%	31.1%	3.99
88460e8ebdecbfecb5f96018339819...	PR	maringa	246	17.1%	47.6%	3.42
1ca7077d890b907f89be8c954a0268...	SP	santana de parna...	108	16.7%	50.9%	2.38
e5a3438891c0bfdb9394643f95273d...	SP	limeira	216	15.7%	17.6%	3.97
cd68562d3f44870c08922d380acae5...	SP	ribeirao preto	122	15.6%	14.8%	4.00
2c9e548be18521d1c43cde1c582c6...	SP	mogi das cruzes	124	15.3%	2.4%	4.10
431af27f296bc6519d890aa5a05fdb11	SP	ribeirao preto	116	13.8%	25.9%	3.78
f7ba60f8c3f99e7ee4042fdef03b70c4	SP	sao bernardo do ...	218	13.8%	0.9%	4.22
8160255418d5aaa7dbdc9f4c64ebd...	SP	ibitinga	380	13.7%	24.5%	3.95
b499c00f28f4b7069ff6550af8c1348a	SP	limeira	146	13.0%	18.5%	4.22
7ad32824caee82087b3e2e5f33b1bf...	SP	ibitinga	147	12.9%	3.4%	3.81
855668e0971d4dfd7bef1b6a4133b...	SP	itatiba	300	12.7%	18.3%	3.83
2e90cb1677d35cfe24eef47d441b7c...	SP	sao paulo	127	12.6%	5.5%	4.12
8f2ce03f928b567e3d56181ae20ae9...	SP	pirituba	151	12.6%	5.3%	4.13
dd7ddc04e1b6c2c614352b383efe2...	SP	sao paulo	121	12.4%	36.4%	3.85
b1fc4f64df5a0e8b6913ab38803c57a9	RJ	sao goncalo	107	12.1%	38.3%	3.82
b2479f944e1b90cf8a5de1bbfde284...	SP	ibitinga	101	11.9%	11.9%	3.49
7aa4334be125fcdd2ba64b3180029f...	SP	laranjal paulista	214	11.7%	36.0%	4.13
c31eff8334d6b3047ed34bebd4d62c...	SP	salto	120	11.7%	26.7%	3.85
7e1fb0a3ebfb01ffb3a7dae98bf3238d	SP	franca	155	11.6%	1.3%	4.15
7d13fca15225358621be4086e1eb0...	SP	ribeirao preto	558	11.5%	14.9%	4.03
b33e7c55446eabf8fe1a42d037ac7d...	SP	pradopolis	149	11.4%	0.7%	4.20
2eb70248d66e0e3ef83659f71b2443...	SP	campinas	187	11.2%	32.1%	2.76
59b22a78efb79a4797979612b885d...	MG	uberlandia	125	11.2%	4.8%	3.96
c3867b4666c7d76867627c2f7fb22e...	SP	guara	228	11.0%	4.4%	4.05
a416b6a846a11724393025641d4ed...	SP	sao paulo	157	10.8%	12.1%	4.19
8b28d096634035667e8263d57ba33...	SP	sao jose dos cam...	139	10.8%	5.8%	4.00

Revenue by product category (R$)

top 12 of 74 categories by revenue



Active sellers

3,095

Sellers with 100+ orders

210

97% of customers never come back — and delivery barely moves that

Repeat purchase is structural on this marketplace; the first order's review score does not predict a return

Month: All | Region: All | State: All

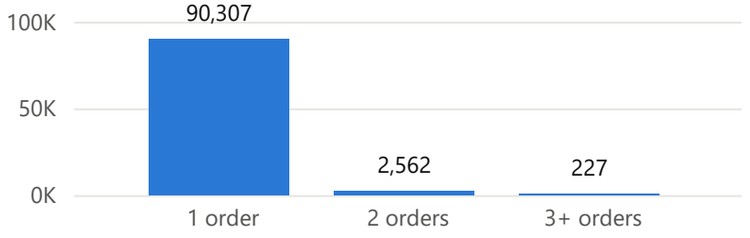
Customers
93,096

Customers who reorder
3.00%

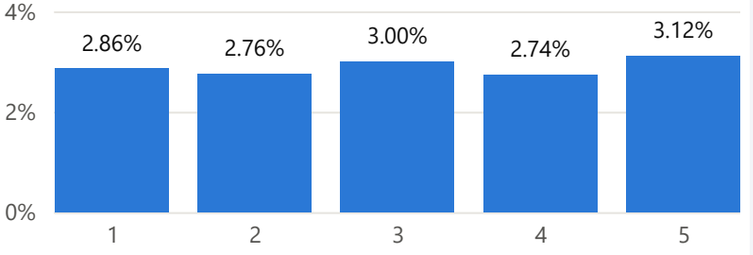
Repeat rate: first order on time
3.03%

Repeat rate: first order late
2.52%

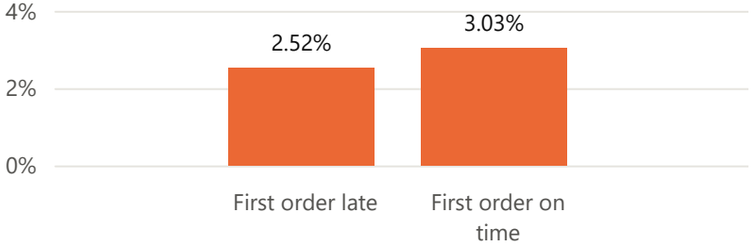
Orders per customer
97.0% place exactly one order



Repeat rate by the first order's review score
flat: a 1-star first order returns as often as a 5-star one



Repeat rate by first-order delivery
3.03% on time vs 2.52% late



Cohort retention: share of each month's new customers who ordered again N months later (2017 cohorts)

rows = acquisition month, columns = months after the first order; darker = more customers came back

Cohort	1	2	3	4	5	6	7	8	9	10	11	12
2017-01	0.28%	0.28%	0.14%	0.42%	0.14%	0.42%	0.14%	0.14%		0.42%	0.14%	0.70%
2017-02	0.18%	0.31%	0.12%	0.43%	0.12%	0.25%	0.18%	0.12%	0.18%	0.12%	0.31%	0.12%
2017-03	0.44%	0.36%	0.40%	0.36%	0.16%	0.16%	0.32%	0.32%	0.08%	0.36%	0.12%	0.20%
2017-04	0.62%	0.22%	0.18%	0.27%	0.27%	0.35%	0.31%	0.31%	0.18%	0.27%	0.09%	0.04%
2017-05	0.46%	0.46%	0.29%	0.29%	0.32%	0.41%	0.14%	0.26%	0.26%	0.26%	0.35%	0.23%
2017-06	0.49%	0.40%	0.43%	0.30%	0.40%	0.36%	0.23%	0.13%	0.20%	0.30%	0.36%	0.16%
2017-07	0.53%	0.35%	0.24%	0.29%	0.21%	0.32%	0.11%	0.19%	0.27%	0.21%	0.29%	0.13%
2017-08	0.69%	0.35%	0.27%	0.35%	0.52%	0.30%	0.27%	0.15%	0.15%	0.25%	0.20%	0.12%
2017-09	0.70%	0.55%	0.27%	0.45%	0.22%	0.22%	0.25%	0.27%	0.20%	0.25%	0.07%	
2017-10	0.72%	0.25%	0.09%	0.23%	0.21%	0.21%	0.37%	0.28%	0.18%	0.21%		
2017-11	0.57%	0.37%	0.17%	0.17%	0.18%	0.11%	0.18%	0.13%	0.06%			
2017-12	0.21%	0.28%	0.34%	0.26%	0.21%	0.17%	0.02%	0.19%				

What this means

Retention is structural: in any month after the first order, only 0.1–0.7% of a cohort buys again, whatever the month.

A late first order moves the repeat rate from 3.03% to 2.52% — real, but a rounding error next to the 97% who never return.

So the business case for fixing delivery is review scores and reputation (pages 1–3), not repeat revenue.